



VISION-BASED DANGER ASSESSMENT AND RISK SCORING IN SECURITY SYSTEMS

Introduction & The Paradigm Shift

Vision-Based Danger Assessment and Risk Scoring in Security Systems

- **Core Theme:** The accurate prediction of danger levels in video content is critical for enhancing public safety, autonomous systems, and security surveillance.
- **The Paradigm Shift:** Transitioning from binary "danger detection" (e.g., detecting a specific weapon or fire) to comprehensive "danger assessment." This involves evaluating the broader context, object-object interactions, and potential human risks over time to assign a quantifiable danger score.

The Foundation - VIDAS Dataset Construction

- **Dataset Purpose:** To provide a standardized benchmark for evaluating how well machine learning and deep learning models perceive and quantify danger.
- **Composition & Metadata:** 100 YouTube videos scraped using the Pytube library. Rich metadata (Video ID, Title, URL, Duration) was compiled to maintain data integrity.
- **Structured Danger Categories:** Risks are categorized into four key areas: extreme sports, accidents, stunts, and workplace hazards/natural disasters.
- **Scale Calibration:** Danger is quantified on a strict 0-10 metric based on inherent relative risk (e.g., beginner rock climbing = 1, wingsuit flying near the ground = 10).
- **Annotation Pipeline:** The VGG Video Annotator tool was utilized to systematically mark precise temporal segments (start and end timestamps) where the heightened danger occurs.

Dataset Statistics

- **Ground Truth Calculation:** 18 human participants evaluated each video. The ground truth for a video's danger rating is mathematically defined as the average of all evaluator ratings.
- **Temporal Averaging:** The precise action boundaries were calculated by averaging the human-annotated start and end timestamps.
- **Variance in Perception:** The standard deviation of ratings measures human consensus. High deviation (e.g., a controlled building demolition) indicates subjective disagreement, while low deviation (e.g., a simple juggling trick) shows strong consensus.
- **Dimensionality Reduction:** A t-SNE graph plotted 2D embeddings of the video summaries, revealing a diverse distribution of scenarios color-coded by their danger severity.

VARs Methodology (Model Architectures)

- **Objective:** To perform a comparative analysis of machine learning and deep learning models to predict danger ratings.
- **Framework 1 (Classification):** Combines visual embeddings from CLIP with textual summary embeddings from GPT. The concatenated data is passed through a Neural Network to classify videos as "High Alert" (≥ 7) or "No Alert" (< 7).
- **Framework 2 (Classification):** Uses only textual video summary embeddings generated by GPT. These are fed into a Support Vector Machine (SVM) using 10-fold cross-validation.
- **Framework 3 (Regression):** Combines visual CLIP embeddings with textual BERT embeddings. A Neural Network performs regression to predict an exact continuous danger score from 0 to 10.

VARs Results & Key Takeaways

- **Framework 1 Performance:** The multimodal CLIP + GPT classification model achieved the highest test set accuracy of 85%.
- **Framework 2 Performance:** The text-only SVM model achieved a 79% cross-validation accuracy. High variance across folds suggests that relying solely on textual features is insufficient for robust classification.
- **Framework 3 Performance:** The regression model successfully predicted precise danger ratings with a Mean Squared Error (MSE) of 0.43.
- **Key Takeaway:** Models that integrate both visual and textual embeddings significantly outperform those relying on a single modality.

VIDAS Methodology (LLM Evaluation & Vision-Language Prompting)

- **Objective:** To evaluate how "human-like" Large Language Models (LLMs) are when assessing danger using the VIDAS dataset.
- **Stage 1 - Video Summarization:** A Large Vision Model (Gemini-1.5-Pro) condensed untrimmed videos into text. The LVM was prompted with a strict questionnaire targeting: the danger source, safety equipment usage, professional involvement, affected humans, and potential aftermath.
- **Stage 2 - LLM Prompting Techniques:**
 - a. **Zero-shot Prompting (ZS):** The LLM rated the summary on a 0-10 scale with zero prior examples, testing its inherent generalization capabilities.
 - b. **Fixed Few-shot Prompting (FFS):** The dataset was sorted by average rating, and every 11th video was injected into the prompt as a predefined example to anchor the rubric.
 - c. **N-shot Prompting:** Using a sentence transformer, the system computed cosine similarity scores to dynamically retrieve and inject the top N most contextually similar video summaries into the prompt.

Multimodal Meta-Evaluation Results

- **Evaluation Metric:** Performance was measured against human ground truth using Mean Squared Error (MSE) to quantify the difference between the LLM's predicted rating and the average human rating.
- **Model Benchmarks:** GPT-4-turbo achieved the most "human-like" perception with the lowest MSE (4.70) under Fixed Few-Shot prompting. Gemini-1.5-Pro outperformed others in Zero-Shot evaluation with an MSE of 11.01. GPT-3.5-turbo struggled across both frameworks.
- **Ablation Study Insights:** An ablation study on N-shot prompting revealed that steadily increasing N (the number of context examples provided) up to 40 consistently decreased the MSE. This proves that expanded contextual examples directly improve evaluation accuracy.

Limitations & Future Architecture

- **LVM Visual Failures:** Large Vision Models occasionally fail to detect subtle threats; for example, missing a snake behind a hammock with a baby until explicitly prompted.
- **The Occlusion Problem:** Objects that are hidden or partially visible limit the accuracy of current LLM evaluations.
- **Dataset Constraints:** The relatively small dataset size (100 videos) may limit generalizability.
- **Future Directions:** Expanding the dataset, fine-tuning domain-specific models, and incorporating temporal analysis (like RNNs) to understand events unfolding over time. Implementing advanced spatial techniques like 3D Gaussian Splatting and Neural Radiance Fields (NeRFs) could resolve occlusion issues.

Overall Conclusion

- **Synthesis:** Combining visual recognition with natural language processing is the most effective way to automate danger assessment.
- **LLM Potential:** Large Language Models show high potential for achieving human-like risk evaluation when provided with appropriate multimodal context and robust prompt engineering.

THANK YOU